

Diagnosing GPS Data Reliability Across Vendors & Providers

Presented by: Mallika Kommera, Xiang Li, Parul
Chaudhary, Isabel O'Grady, Zhichen Yang



4MATIV aims to validate GPS reliability to enable accurate route monitoring and actionable insights

Situation

4MATIV operates TOMS, collecting GPS data from **multiple vendors** to track school buses. However, inconsistent GPS data **creates uncertainty** in monitoring, decision-making, and vendor accountability

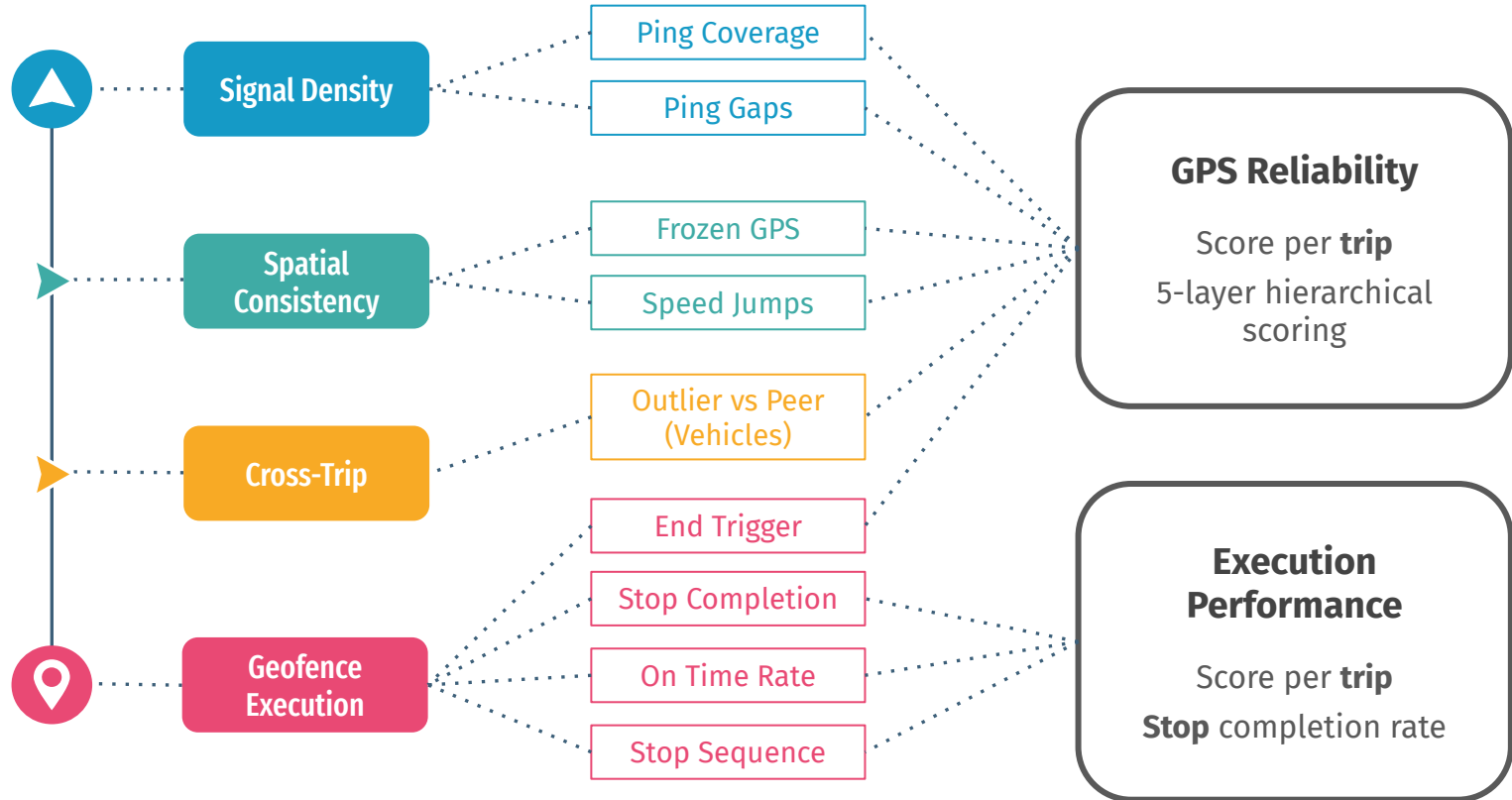
Challenge

GPS reliability **varies across vendors and GPS providers**, making it difficult to determine whether issues are due to actual **route execution** (missed stops, delays) or **data quality** problems. We wanted to find out: Which vendors and providers are consistently having GPS reliability issues.

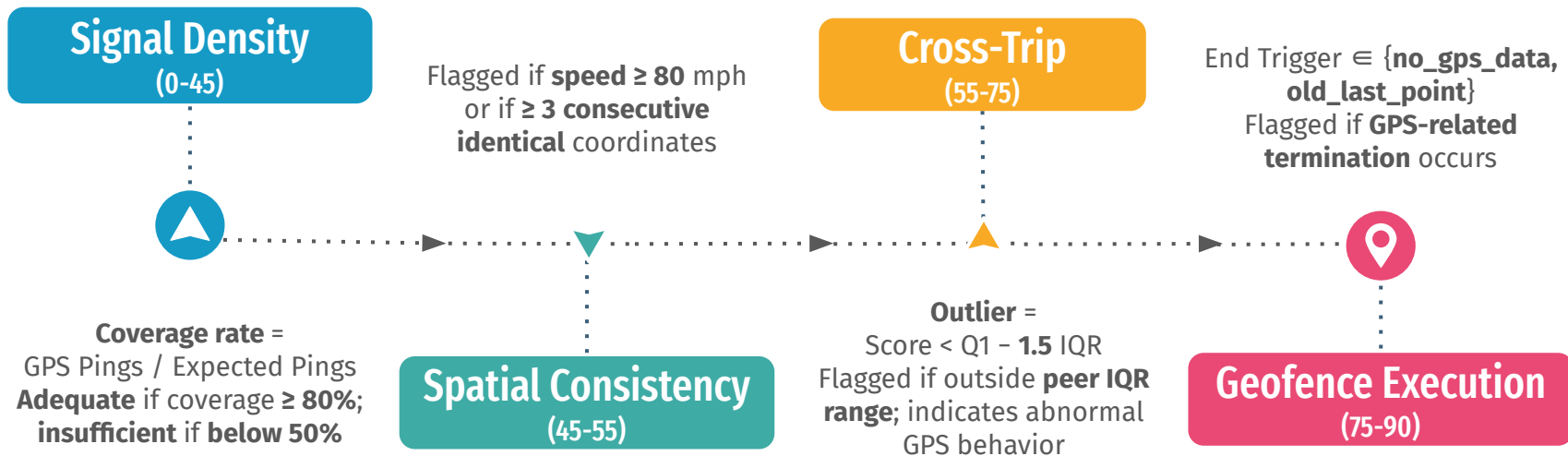
Outcome

To address this, we wanted to find a reliable way to validate GPS data. We computed a **GPS reliability** and **route execution** score and developed a dashboard that displays both scores. The dashboard can help identify **data vs. execution** issues on a vendor and provider-level allowing for better monitoring.

Our Approach to Calculating GPS Reliability and Execution Scores



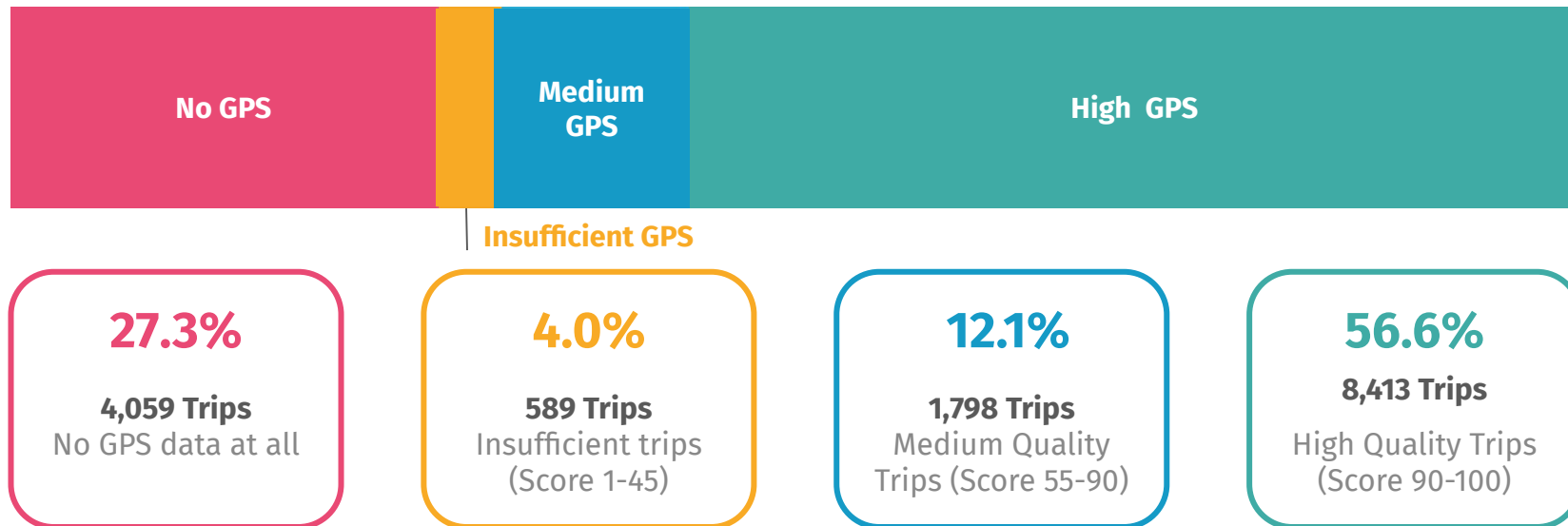
GPS Reliability Is Driven by Data Coverage, Signal Integrity, and Consistency Across Trips



* Scores (90-100) act as an overall consistency adjustment, scaled by GPS coverage quality.

GPS Reliability Distribution Shows Data Quality Gap

14,859 trips · 2.3 million pings · 16 vendors · 5 GPS providers · Sep 2025 - Feb 2026



What does this distribution mean for 4Mativ?

4MATIV cannot validate whether 1 in 3 routes ran as planned. Any operational decisions on these trips is a guess.

*Scores between 45-55 were classified as low, but no trip fell within that category

Poor Coverage and Frequent Gaps Leave 4Mativ Operating Blind

Ping Coverage (Quantity)

- 49.3% average coverage rate
 - Half of expected pings are missing. Only 4.7% of trips report correct ping coverage.
- Worst: iCabbi — 30.5% avg coverage
 - 60s polling interval — a contractual issue, not a technical one.

Ping Gaps

- 56.4% of trips have at least 1 large gap
- 31.8% of trips have 10+ large gaps
- 130 min worst single gap recorded
- Average max gap per trip: 248 seconds (~4 min)

Provider patterns:

- **Synovia:** 73% of trips with gaps, avg 9.4 gaps/trip — constant dropout
- **Samsara:** 64.7% with gaps across 7,847 trips — largest business impact
- **iCabbi:** Only 4.1% gap rate shows consistent underreporting, not dropout

So what?

A gap means the bus disappeared. Stops visited during that window cannot be confirmed — the route is not validatable for that period.

GPS Freezes and Teleports Reveal Distinct Provider Failures & Cross-Trip Consistency Shows Fixable Problems

Frozen GPS

9.6%

of trips have frozen GPS
1,427 trips exceed 10% threshold

98.1%

worst single trip frozen ratio – almost
entire trip was a ghost location

Worst Providers:

Icabbi – 22.9%
OSG – 11.3%

Speed Jumps

2.5%

of trips have impossible speed jumps
with avg 20.6 jumps per affected trip

932

jumps in single worst trip – GPS trace
is completely meaningless

Worst Providers:

Synovia – 19.5%
Icabbi – 3.3%

Cross-Trip & Geofence

144 trips

Flagged as outliers
1% of trips – isolated device failures
not provider or route problems

0 trips

GPS-terminated
No trips were lost due to system
failure

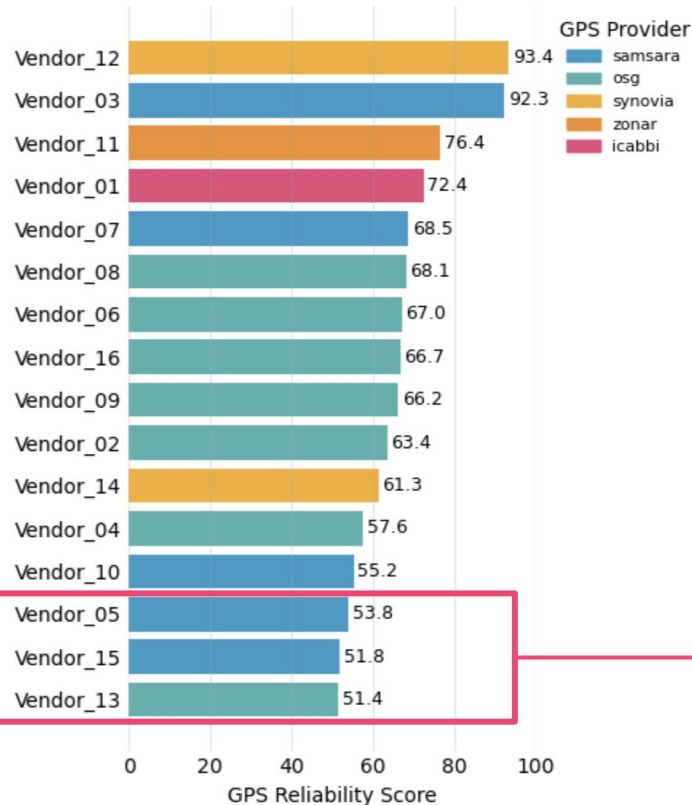
GPS health at stop level:

- **Healthy:** 23% of stop visits
- **Intermittent:** 10% of stop visits
- **Severe gap:** 67% of stop visits

Key Insight: Systemic API issues may have improved with final test, but device-level failure remain widespread. 144 outlier trips point to specific hardware problems that are isolated enough to fix.

Missing and Frozen GPS Drive Low GPS Reliability Across Vendors

Vendor GPS Reliability (by GPS Provider)



General Findings

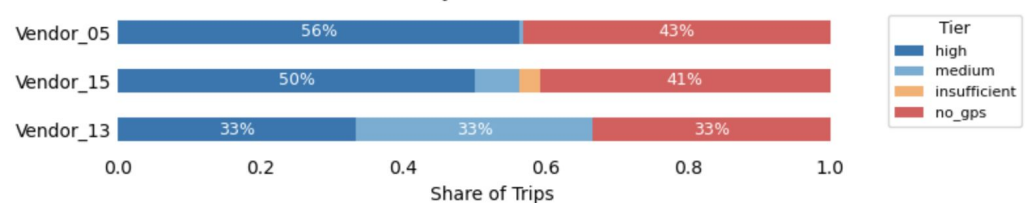
- ~4,000 trips (27%) lack GPS data, making validation impossible
- Reliability varies widely across vendors → **uneven GPS confidence**



Vendor-level findings

- Poor performance driven by **high “no GPS” rates** and **frozen GPS pings**
- Indicates **systemic issues** (device/provider) rather than random noise

Reliability Tier Distribution



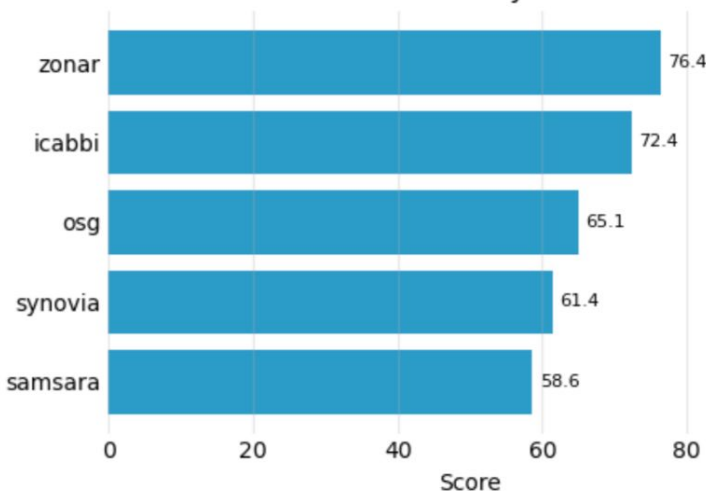
Missing and Insufficient Data Drive Low GPS Reliability Across GPS Providers



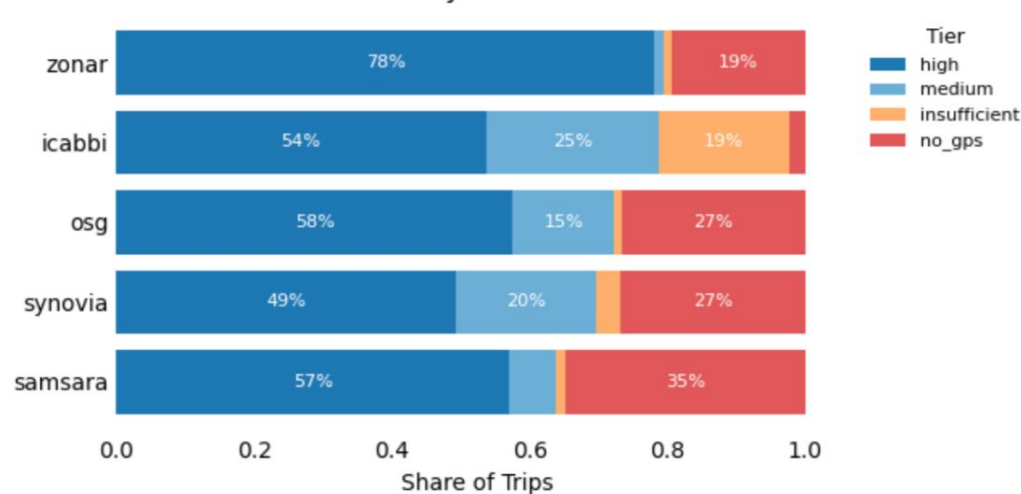
GPS Provider-level findings

- **Samsara** and **Synovia** has the lowest GPS reliability, driven by **high “no GPS”** and **low GPS coverage rates**, indicating systemic data issues.

GPS Reliability Score



Reliability Distribution



Execution Score: Coverage & Key Metrics

80.2

Average Execution
Score

Execution Score = Stop
Completion (40%) + On-Time Rate
(40%) + Stop Sequence (20%)

86.6%

Average Stop
Completion Rate

Stop Completion Rate =
Completed Stops / Planned Stops

74.9%

Average On-Time
Rate

On-Time Rate = Stops within ± 10
minutes / Completed Stops

83.4%

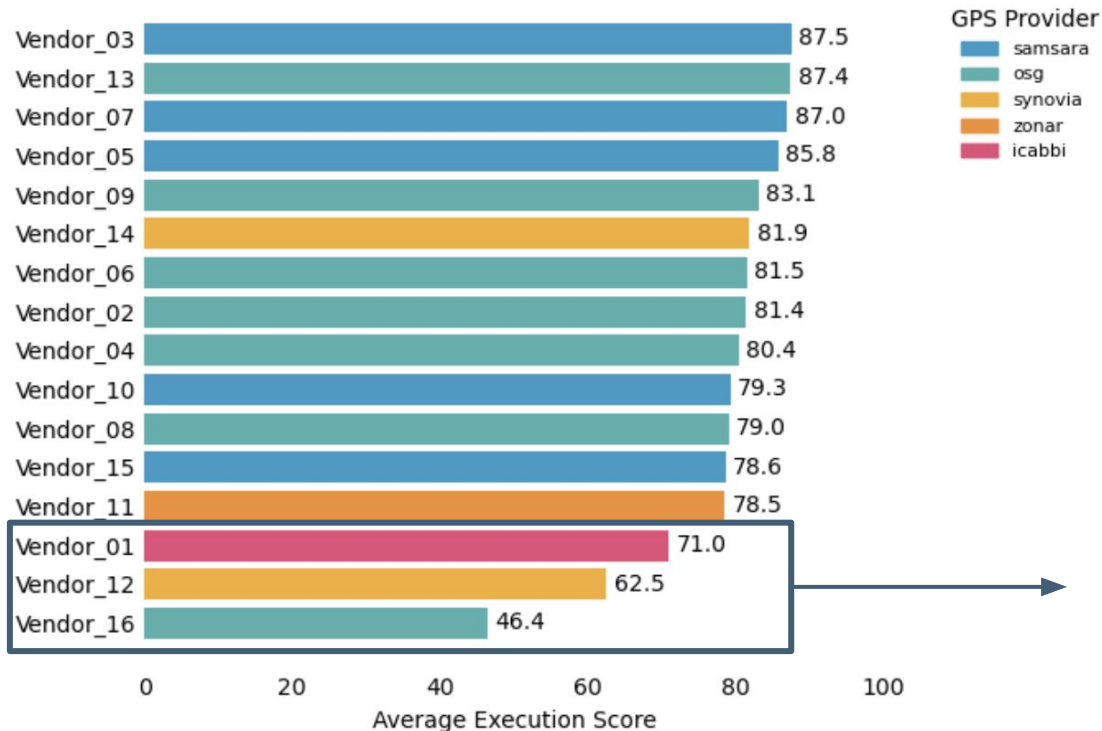
Average Sequence
Adherence

Sequence Adherence = Stops in
correct order / Total stops
movement



On-Time Rate: Key Differentiator Across Vendors

Average Execution Score by Vendor



Why are these vendors underperforming?

Vendor_01

Score: 71.0



Vendor_12

Score: 62.5



Vendor_16

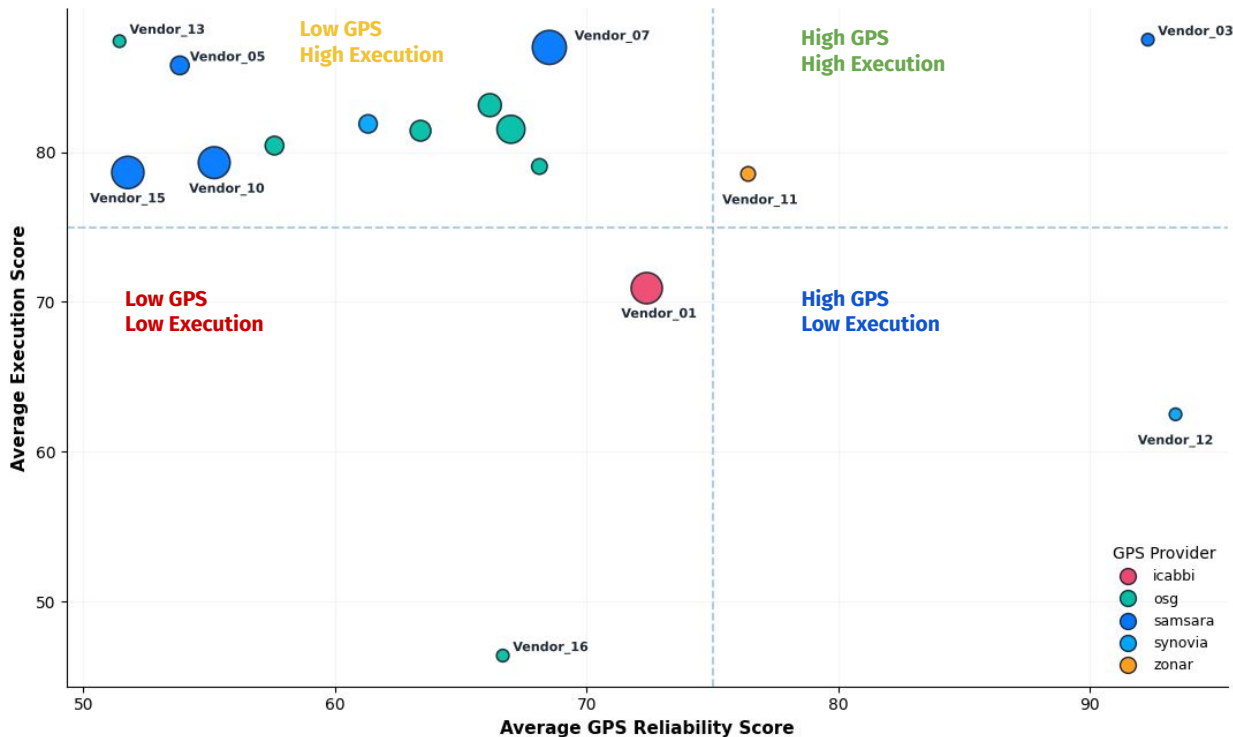
Score: 46.4



* One caveat: Hierarchical clustering shows vehicle type (bus, van, cab) drives vendor performance differences in on-time rate.

GPS Reliability v/s. Average Execution: Analysis & Interpretation

Vendor Segmentation: GPS Reliability vs Execution



Quadrant Interpretation

High GPS + High Execution

Reliable data AND good route execution. No action needed.

e.g. Vendor_03, vendor_11

Low GPS + High Execution

Route likely executed well but GPS data limits confidence. Prioritize GPS device investigation.

e.g. Vendor_13, Vendor_10, Vendor_15

High GPS + Low Execution

GPS data is reliable — execution problems are real. Investigate operational issues.

e.g. Vendor_12

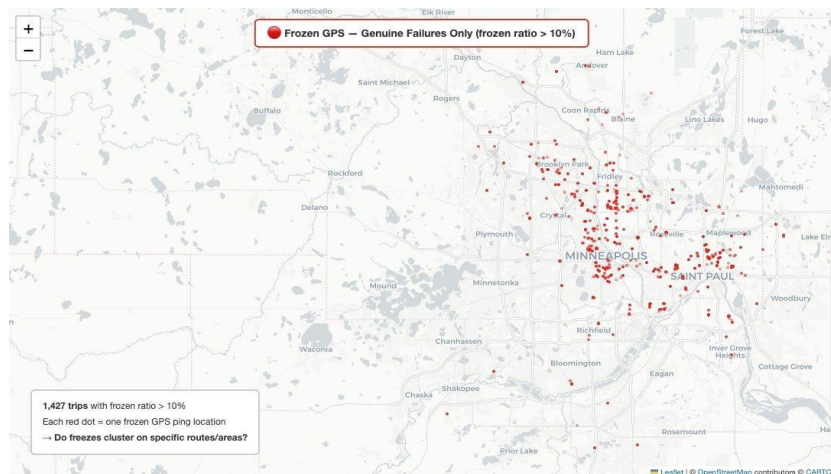
Low GPS + Low Execution

Cannot distinguish GPS failure from execution failure. Fix GPS first.

e.g. Vendor_01, Vendor_16

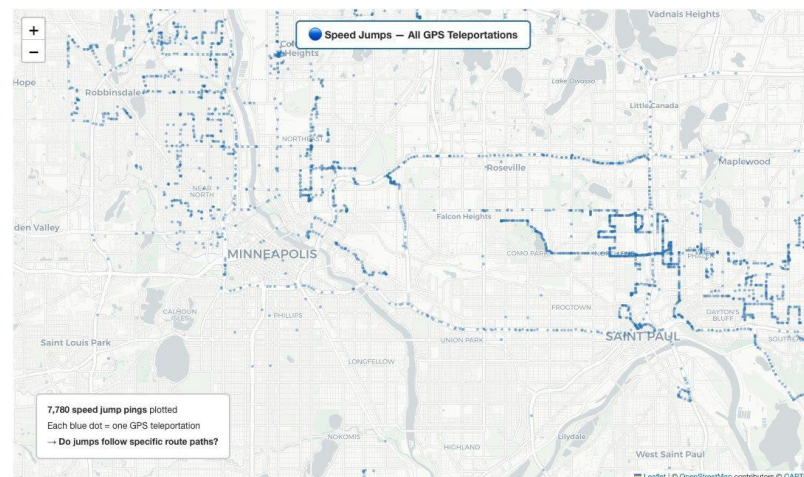
Geographical Analysis: Where GPS Failures Happen Reveals Why They Happen

Frozen GPS — Scattered (Device Failure)



Finding: Freezes are randomly scattered — no route pattern. Points to device hardware failure on specific buses (iCabbi/OSG).

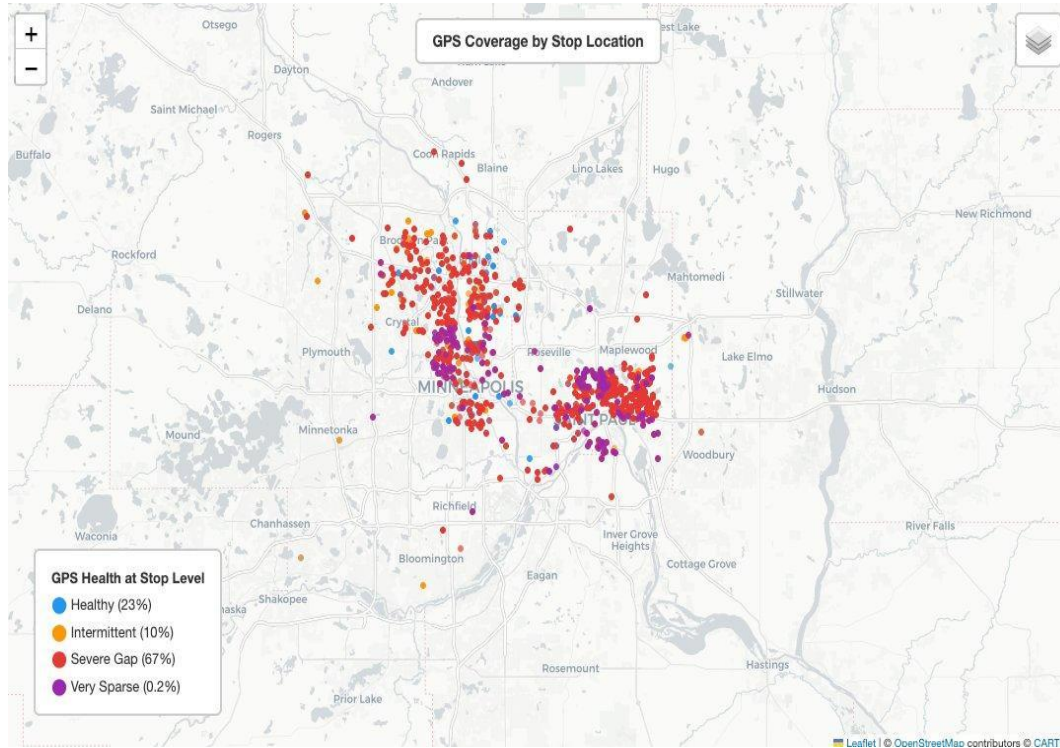
Speed Jumps — Route Corridors (API Failure)



Finding: Speed jumps trace exact route corridors. GPS teleports along known paths. Provider API issue (Samsara/Synovia).

Spatial Correlation Analysis confirms: areas with frozen GPS also tend to have poor stop coverage — same underlying signal problem. Speed jumps occur independently — a separate API issue on specific route corridors. Two failures, two different fixes.

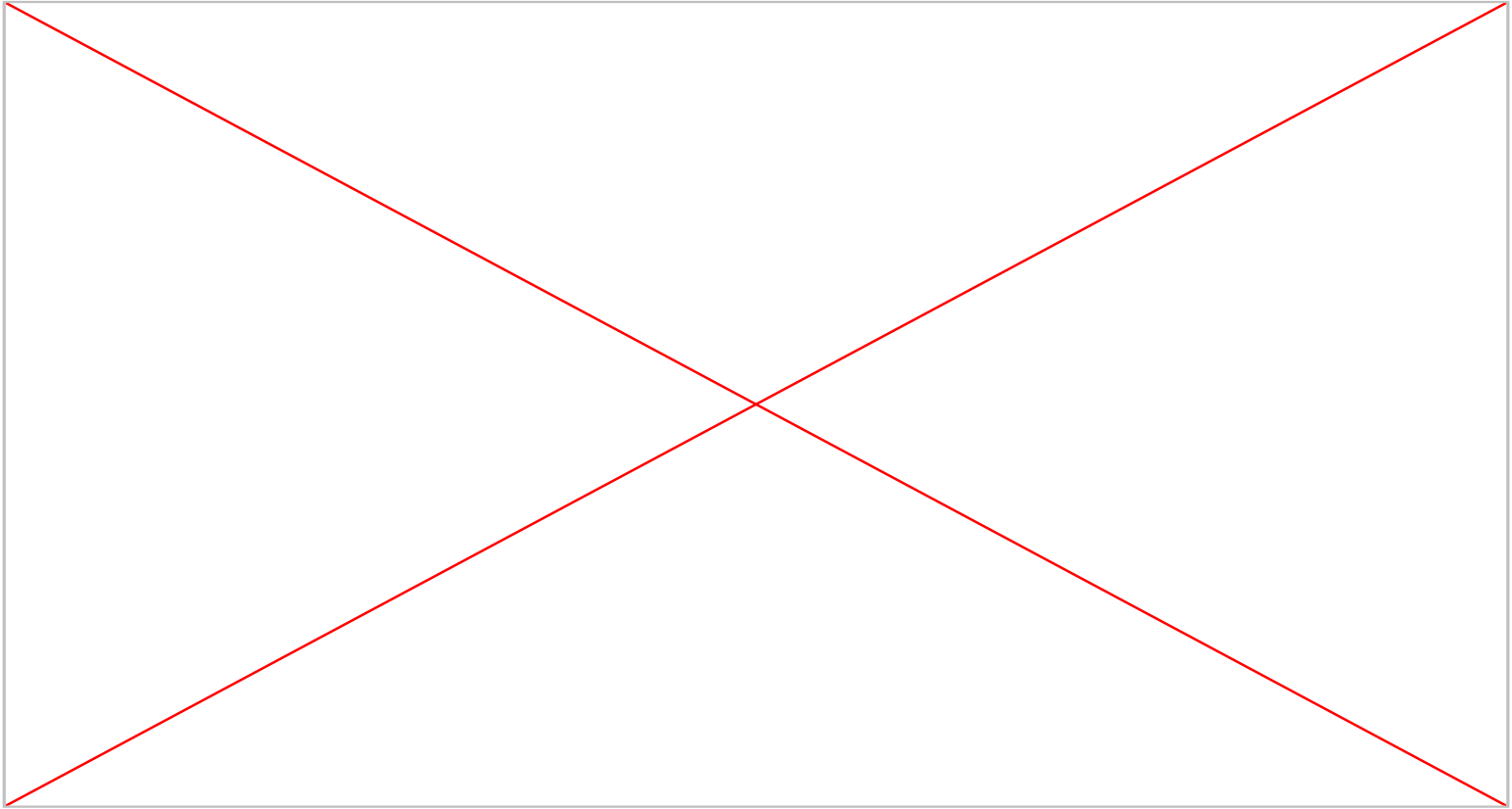
GPS Coverage by Stop Location: Red Dominates Everywhere



Key findings:

- Red stops are spread across the entire metro — no neighbourhood has consistently healthy GPS
- Very sparse — clusters in Saint Paul east side
- 4MATIV cannot validate stop visits with confidence in any zone of the city
- Statistically confirmed: areas with frozen GPS also have more severe gap stops ($r = 0.49$) — same underlying signal problem

Dashboard Demonstration



Recommendation 1 - Investigate Provider Issues

Key Insight

- We were able to identify key limitations of each provider.
 - Icabbi: Frozen GPS & Speed Jumps
 - Samsara: Ping Gaps & Speed Jumps
 - Synovia: Ping Gaps
 - OSG: Frozen Signals

We Suggest

- Investigate each providers failures at the device and geography level to identify if there are patterns that point towards more isolated issues over systemic ones.
- Ideally, providers will be able to solve their API and systemic issues along with the fix the absence of data for trips.

Recommendation 2 - Audit the 144 Flagged Devices Before the Next Service Period

Key Insight

- Coverage is a systemic issue by provider, but quality seems to be a device level failure. 144 trips were flagged as isolated device level failures.
 - Peer group = same route_id + trip_type
 - This controls for route difficulty → only flags genuine outliers

We Suggest

- Cross-reference flagged trips to identify which physical devices appear most frequently across flagged trips
- Replace or reconfigure devices – A targeted swap or update on a small number of devices could eliminate the majority of outliers without any system-wide intervention

Appendix

Appendix 1a – Spatial Consistency : Two Distinct Spatial Failure Modes Across Providers identified by LOF

540 outlier trips (5%)

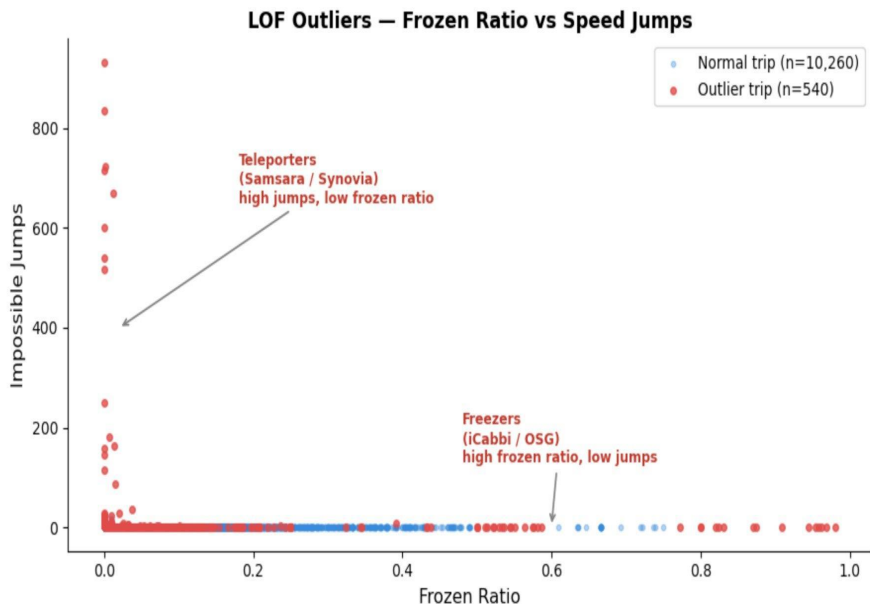
Anomalous vs their peer trips

Freezers — iCabbi + OSG

GPS locks onto one coordinate

Teleporters — Samsara + Synovia

GPS jumps to wrong coordinates



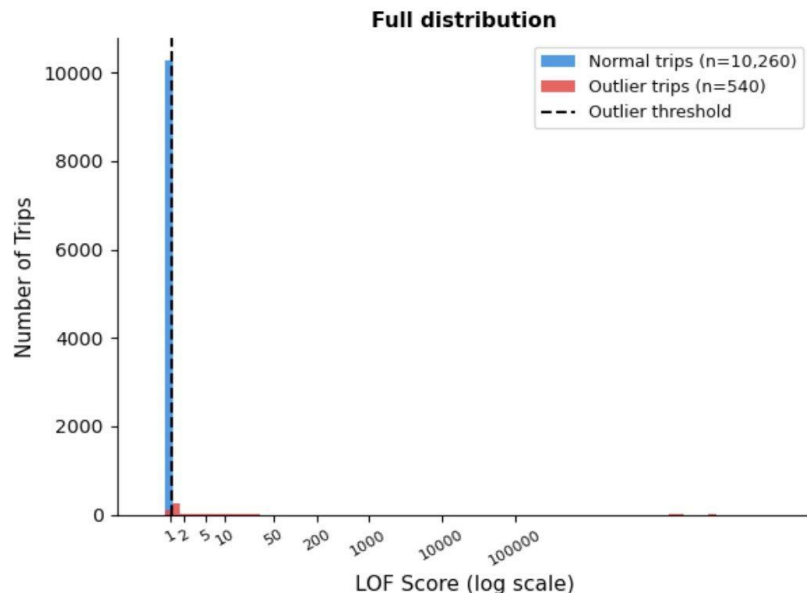
Key Inference from the LOF Analysis:

- Red = outlier trips.
- Top-left cluster = teleporters (high jumps).
- Bottom-right scatter = freezers (high frozen ratio).

Appendix 1b - LOF Score Distribution: Outlier Trips Are Genuinely Extreme, Not Random Noise

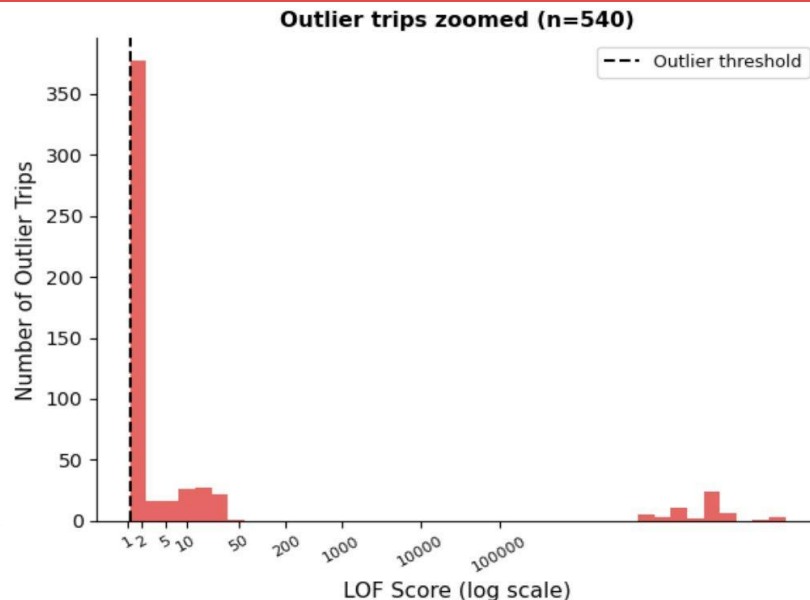
Full Distribution

10,260 normal + 540 outlier trips



Outlier Trips Zoomed

540 outlier trips — spread across extreme LOF scores



Blue = normal trips (score ~1). Red = outliers with scores up to 100,000+ — these are real device failures, not statistical noise.

Appendix 2a – Methodology – Vendor Performance Segmentation

Objective

Evaluate vendor performance by comparing **GPS reliability** and **execution quality**, and identify patterns by **GPS provider** while accounting for **trip volume**.

1) Data Preparation

- a) Aggregated trip-level data to **vendor level**
- b) Key Metrics:
 - i) **GPS Reliability** (avg signal quality)
 - ii) **Execution Score** (route performance)
 - iii) **Trip Volume** (used as confidence indicator)

2) Visualization Design

- a) **X-axis:** GPS Reliability
- b) **Y-axis:** Execution Score
- c) **Color:** GPS provider
- d) **Bubble size:** scaled by trip volume

3) Segmentation Approach

- a) Fixed thresholds for interpretability:
 - i) High GPS ≥ 75 , High Execution ≥ 75
- b) Enables consistent quadrant comparison across vendors

Presentation Choices:

- Labeled **key vendors only** (outliers + high-impact)
- Manual label placement for clarity
- Low-volume vendors treated as **potential outliers**

Key Note:

- Bubble size reflects **data confidence**, not performance
- Color enables **provider comparison**, not clustering
- Additional analysis (clustering, PCA) used to validate segmentation patterns

Appendix 2b: Vendor-Level Performance Summary

Vendor-level summary used as the foundation for segmentation

GPS Reliability vs Execution Score (Vendor Level)

vendor_label	gps_provider	avg_gps_reliability	avg_execution_score	total_trips
Vendor_13	osg	51.433333	87.400000	3
Vendor_15	samsara	51.764686	78.640796	2373
Vendor_05	samsara	53.831041	85.784775	509
Vendor_10	samsara	55.197947	79.295101	2289
Vendor_04	osg	57.592427	80.432764	515
Vendor_14	synovia	61.312371	81.881009	485
Vendor_02	osg	63.396332	81.423552	736
Vendor_09	osg	66.151848	83.129379	1001
Vendor_16	osg	66.666667	46.400000	3
Vendor_06	osg	66.989518	81.521573	1679
Vendor_08	osg	68.122500	79.040223	240
Vendor_07	samsara	68.515969	86.985618	674
Vendor_01	icabbi	72.374099	70.954767	2193
Vendor_11	zonar	76.388387	78.546341	155
Vendor_03	samsara	92.300000	87.500000	2
Vendor_12	synovia	93.400000	62.500000	2

Appendix 3 - Why Do Vendors Differ? Fleet Composition Is the Answer

Hierarchical clustering on performance + vehicle composition

Cluster 1: On-time rate is the weak link

Van-Only Vendors

Vendor_04 · Vendor_08 · Vendor_10 · Vendor_15

79.4% Execution Score

73.2% On-Time Rate

88.1% Completion

GPS Reliability: 88.3

100% van fleet. Completion is decent but on-time rate lags Cluster 2 by ~8 pts.

Cluster 2: Best overall performance

Bus-Dominant Vendors

Vendor_02 · Vendor_05 · Vendor_06 · Vendor_07 ·
Vendor_09 · Vendor_11 · Vendor_14

82.8% Execution Score

80.8% On-Time Rate

86.9% Completion

GPS Reliability: 91.8

97% bus fleet, highest GPS reliability (91.8). Overall the best execution performance cluster.

Cluster 3: Structural outlier

Cab-Dominant Vendor

Vendor_01

71.0% Execution Score

59.8% On-Time Rate

79.4% Completion

GPS Reliability: 84.1

87.5% cab fleet. On-time rate 20+ pts below Cluster 2. High sequence score (92.7%) due to having fewer stops.

**Thank
You!**